

A Multi-Vendor Corpus of Reference Low-Dose CT Reconstructions with Calibrated Per-Pixel Uncertainty and Signal-Equivalence Credentials

[TODO: First A. Author]^{*1,2}, [TODO: Second B. Author]¹, and [TODO: Third C. Author]²

¹University of Texas Southwestern Medical Center, Dallas, TX, USA

²PWM Protocol Foundation

Working draft v0.3 (Data Descriptor) — September 11, 2026

Abstract

Low-dose CT reconstruction research is rich in methods but poor in *shared, reusable derived data*: published methods release a model or a number, rarely the per-case reconstructed volumes, the per-pixel uncertainty that a clinical user would triage on, or a machine-checkable certificate of diagnostic equivalence. We describe a deposited corpus that supplies exactly this layer on top of the multi-vendor PWM-LDCT benchmark ([to be confirmed](#), 2027a). For every test scan, at four dose levels (10%, 25%, 50%, 100%) and across multiple CT vendors, the corpus provides: (i) a **reference reconstruction** produced by an open, fully specified unrolled-iterative pipeline; (ii) a **per-pixel uncertainty map** from a five-model deep ensemble; (iii) the **paired per-pixel error map** against the full-dose reference, enabling calibration studies without re-running any model; (iv) **downstream-task output maps** (lung-nodule detection scores); (v) machine-readable **signal-equivalence credentials** computed under the framework of [to be confirmed](#) (2027b); and (vi) the matched reconstructions of five baseline methods for comparison. Every record is content-addressed, released under CC BY 4.0, and accompanied by an Apache-2.0 generating pipeline packaged as a pinned Docker RunBundle so the corpus is regenerable bit-for-bit within floating-point tolerance. Technical validation reports reconstruction fidelity, cross-vendor consistency, and—centrally for reuse—the agreement between the released uncertainty maps and the released error maps (Spearman ρ). The corpus is intended for reuse in uncertainty-calibration research, cross-vendor robustness studies, downstream-task evaluation under dose reduction, and as a stable comparison baseline for new reconstruction methods.

1 Background & Summary

Deep-learning low-dose CT reconstruction has produced a decade of method papers reporting fractional-dB gains (Chen et al., 2017; Wang et al., 2020; Song et al., 2022), yet the field shares almost no *derived data*. Two structural gaps follow. First, reconstructed volumes are seldom released: a researcher who wants to study how reconstruction error distributes across anatomy, dose, or vendor must retrain each method from scratch. Second, the two quantities most relevant to clinical deployment—*per-pixel uncertainty* and a *certificate of task-level equivalence*—are almost never published as data a third party can inspect. A deployer choosing a reconstruction for a new scanner, or a calibration researcher studying whether predicted variance tracks true error, has nothing reusable to build on.

*Corresponding author: [TODO: email@utsouthwestern.edu]; ORCID [TODO: 0000-0000-0000-0000]. A registered ORCID for *every* author is mandatory for *Scientific Data*; register and list all before submission.

The raw substrate for closing these gaps now exists: the PWM-LDCT benchmark (to be confirmed, 2027a) provides multi-vendor paired (full-dose, low-dose) CT, extending the field’s two single-vendor real-paired resources, AAPM-2016 (AAPM, 2016) and Mayo LDCT-PD (McCollough et al., 2017), both Siemens-only. What has been missing is the *derived layer*: a uniformly produced set of reconstructions, uncertainty maps, error maps, task outputs, and equivalence credentials over that substrate.

This Data Descriptor releases that layer. We generate every record with a single, fully specified open pipeline—an unrolled iterative reconstruction with a weight-shared denoiser and a five-model deep ensemble for uncertainty (Methods)—and deposit the outputs as a content-addressed corpus. We emphasise that the pipeline is a *means of generating reproducible data*, not the object of this paper; its architecture is reported in full only so the data are reproducible and the uncertainty maps are interpretable.

Relationship to the PWM-LDCT benchmark (WS-1). This corpus is deliberately disjoint from the underlying benchmark dataset. WS-1 releases the *raw paired CT scans*; the present corpus releases the *reconstruction, uncertainty, error, task-output, and credential records derived from them*. The two are intended to be cited together: WS-1 as the input data, this corpus as the reference derived data.

Reuse value. The corpus enables, without any model retraining: (1) **uncertainty-calibration research**—the paired (uncertainty map, error map) records are a ready-made benchmark for calibration metrics and recalibration methods; (2) **cross-vendor robustness studies**—reconstructions and errors are vendor-labelled, supporting leave-one-vendor-out analyses; (3) **downstream-task evaluation under dose reduction**—released task-score maps let others study how dose affects detection without re-deriving reconstructions; (4) **credential-based comparison**—the machine-readable signal-equivalence credentials are a stable target that new methods can report against under a content-addressed framework hash; and (5) a **stable comparison baseline**—five matched baseline reconstructions accompany the reference so new methods compare against identical inputs.

Summary of what is released. Table 1 summarises the record types; section 3 gives the full file-level manifest. All records, for all scans, at all dose levels and vendors, are deposited together under a single repository DOI ([**TODO: figshare/Zenodo DOI on deposit**]).

2 Methods

This section specifies how each released record was generated. It is included to make the corpus reproducible and the uncertainty maps interpretable; it is not a methodological contribution of this Data Descriptor.

2.1 Input data and provenance

Inputs are the PWM-LDCT (to be confirmed, 2027a) scans at signal ratios $r \in \{0.10, 0.25, 0.50\}$, patient-level split into train/validation/test, vendor-labelled. Reference (target) images are the patient’s full-dose reconstruction. No scan crosses the train/test boundary at the patient level; all released records are for the held-out test split.

Low-dose provenance (real-paired vs simulated). Each (vendor, dose) stratum is one of two kinds, recorded per record in `scan_meta.json` (`low_dose_origin` $\in \{\text{real_paired}, \text{simulated}\}$): *real-paired* acquisitions are genuine repeat low-dose scans from the source benchmark (to be confirmed, 2027a); *simulated* low-dose is generated from the full-dose projections by Poisson photon-count reduction at the stated r , as documented in the WS-1 source descriptor (to be confirmed, 2027a); these images are inherited from that release and are not regenerated by the pipeline released here. Reusers must not treat the two as interchangeable: simulated strata characterise algorithmic behaviour under an idealised noise model, real-paired strata characterise behaviour under true acquisition physics. The per-stratum breakdown is reported in Table 3 and carried in every record’s metadata.

The $r = 1.00$ level. Full dose is the reference, not a credentialed dose level. Where an $r = 1.00$ reconstruction is released it is the reference image x^{ref} itself; no error map is emitted at $r = 1.00$ (the error against the reference is identically zero) and no credential is defined there (signal-equivalence credentials compare a *reduced*-dose method against the full-dose reference). All released error maps, task-score maps, and credentials are therefore at $r \in \{0.10, 0.25, 0.50\}$.

Ethics and human-subjects provenance. This corpus contains no new human-subjects data: every record is a derived image or a numerical summary computed from the PWM-LDCT benchmark ([to be confirmed, 2027a](#)), which governs the underlying acquisitions, their institutional-review-board approvals, and informed-consent basis. The specific governing approvals — **[TODO: name the WS-1 IRB protocol number(s) and the Mayo LDCT-PD / AAPM 2016 data-use agreement reference]** — are stated in the WS-1 source descriptor and restated here at submission. The released NIfTI volumes carry no protected health information: voxel arrays only, with all DICOM patient-identifying header fields removed at generation, and `scan_id` is the WS-1 pseudonymous identifier (never an institutional medical-record number). Redistribution follows the per-source basis of the WS-1 release, set out under *Licensing and redistribution* ([section 3](#)); reusers remain bound by the source data-use terms, in particular for any attempt to re-link records to source acquisitions. No date-of-service, geographic, or demographic field finer than the anatomy/vendor labels in `scan_meta.json` is released.

2.2 Reconstruction pipeline

Reconstructions are produced by an unrolled iterative scheme ([Adler and Öktem, 2018](#); [Mardani et al., 2018](#)). Let $\mathcal{R} : \mathbb{R}^{H \times W} \rightarrow \mathbb{R}^{n_{\text{views}} \times n_{\text{dets}}}$ be the Radon forward operator and $\mathcal{R}^\dagger$ its filtered back-projection adjoint. Both are shipped with the released pipeline as a self-contained, differentiable parallel-beam implementation, written as a drop-in stand-in for the `pwm_core.contrib.modalities.ct_radon` interface ([PWM Protocol, 2026](#)); that package is not vendored into this repository, so the operator pinned in the RunBundle is the one that produced every record released here. The adjoint is obtained by automatic differentiation through the forward operator, so the data-consistency term is exact for the operator actually shipped.

Given a low-dose sinogram y , the reconstruction $\hat{x}$ is computed by $K = 10$ data-consistency / denoising iterations ([Algorithm 1](#)). The denoiser f_θ is a U-Net (≈ 4 M parameters, four encoder / four decoder stages with (32, 64, 128, 256) channels, GELU, GroupNorm) whose weights are shared across iterations; the step size τ is a learned scalar initialised at $0.5 / \|\mathcal{R}\|_2^2$.

Algorithm 1 Unrolled iterative reconstruction (per ensemble member)

```

1:  $x^{(0)} \leftarrow \text{TV-FBP}(y)$  ▷ TV-regularized warm start
2: for  $k = 0, 1, \dots, K-1$  do
3:    $x^{(k+1/2)} \leftarrow x^{(k)} - \tau \cdot \mathcal{R}^\dagger(\mathcal{R}(x^{(k)}) - y)$  ▷ Data consistency
4:    $x^{(k+1)} \leftarrow f_\theta(x^{(k+1/2)})$  ▷ Learned denoising
5: end for
6: return  $\hat{x} \leftarrow x^{(K)}$ 

```

2.3 Uncertainty quantification

Each released reconstruction is the mean of a five-model deep ensemble ([Lakshminarayanan et al., 2017](#)): five instances of the architecture above, identical except for the random seed governing weight initialisation and data ordering (seeds {42, 43, 44, 45, 46}). For a low-dose input y , member m produces $\hat{x}_m(y)$; the released point estimate and per-pixel uncertainty are

$$\hat{x}(y) = \frac{1}{5} \sum_{m=1}^5 \hat{x}_m(y), \quad \sigma(y)_{ij} = \sqrt{\frac{1}{5} \sum_{m=1}^5 (\hat{x}_m(y)_{ij} - \hat{x}(y)_{ij})^2}. \quad (1)$$

Deep ensembles are chosen over MC dropout because they calibrate better under distribution shift ([Ovadia et al., 2019](#); [Gal and Ghahramani, 2016](#)) and have bounded ($5\times$) inference cost. The released $\sigma(y)$ map is

the quantity intended for case-level triage; its *validity* as a triage signal is established in Technical Validation by comparison with the released error map.

2.4 Derived per-record quantities

For each test scan and dose level the pipeline emits, in addition to $\hat{x}(y)$ and $\sigma(y)$: the **per-pixel error map** $e(y)_{ij} = |\hat{x}(y)_{ij} - x_{ij}^{\text{ref}}|$ against the full-dose reference, and a **lung-nodule detection score map** from a frozen pre-trained detector applied to $\hat{x}(y)$ (detector specified in Section 2.5). (Version 1 of the corpus releases this single chest / lung-nodule task; additional tasks such as abdominal liver-lesion segmentation are deferred to a future release so that the released records and the task description remain in exact correspondence.) In addition, at the *cohort* level—one record per (task, r , subpopulation, method) stratum rather than per scan—the pipeline emits a **signal-equivalence credential** (5-tuple $(r, T, \varepsilon, \alpha, \Pi)$ with verdict and confidence interval) computed with the `pwm_dose_equivalence` library (to be confirmed, 2027b); each credential is internally audited with `pwm-audit` before it is written, and is regenerable from the released task-score arrays. The five baseline methods (Section 3) are run through the identical emit path so their records are field-for-field comparable.

2.5 Frozen downstream-task detector

The lung-nodule detection score maps are produced by a single *frozen* detector held fixed across every method, dose level, and record, so the task layer measures the diagnostic content each reconstruction preserves rather than any adaptation of the detector itself. The detector is **nnDetection** (Baumgartner et al., 2021)—a self-configuring 3D Retina U-Net—run in its published configuration without modification; the exact release is pinned in Supplementary Table S2. It is trained *once* on the **LUNA16** benchmark (Setio et al., 2017), the 888-scan LIDC/IDRI (Armato III et al., 2011) subset annotated for pulmonary nodules with $\geq 3/4$ radiologist agreement, and is never fine-tuned on PWM-LDCT reconstructions. The LUNA16 training patients are disjoint from the PWM-LDCT cohort, so the detector is an external, leakage-free evaluator applied identically to all methods.

Operating-point provenance. The credentialing operating point is fixed *before* any PWM-LDCT reconstruction is scored: on the held-out LUNA16 validation fold we take the detection-score threshold whose free-response ROC (FROC) yields a mean of 0.1 false-positive detections per scan ($\text{FPR} = 0.1$) and freeze it for all subsequent scoring. The released task-score maps are the raw, pre-threshold detector scores, so reusers may re-derive performance at any operating point; the AUC in Table 9 and the task credentials are read at this single frozen $\text{FPR} = 0.1$ point, identical across all methods and doses. The frozen weights, inference configuration, and version pin are deposited in the RunBundle (Section 6) so the task layer is independently reproducible.

2.6 Training and generation configuration

Models are trained on the PWM-LDCT training split with paired targets at $r \in \{0.10, 0.25, 0.50\}$ mixed per minibatch with weights (0.4, 0.4, 0.2); augmentation is physical-consistency-preserving only (random flip $p=0.5$; rotation $[-5^\circ, 5^\circ]$; window-level shift $[-50, 50]$ HU). Optimiser AdamW (Loshchilov and Hutter, 2019), cosine LR with 5% warmup, batch size 4 at 512×512 , single A100; CuDNN deterministic, all RNGs seed-propagated. The complete hyperparameter table, library version pins, and compute disclosure accompany the deposit (Supplementary Tables S1–S3) so the corpus is regenerable.

Training status. As of 2026-08-17, the method scaffold is complete and CPU-verified (19 tests; `pwm-recon smoke` runs the full Phase-3 pipeline on synthetic data), but the production deep-ensemble training described above has **not yet been executed**: the WS-1 v0.5 harmonized data tree is not yet landed and no data-center GPU has been booked (current workstation: RTX 4060 Laptop 8 GB, insufficient; planned envelope ≈ 600 –1,800 A100-h per `COMPUTE_RESOURCES.md`). All quantitative Technical-Validation tables and figures below therefore remain **[TODO:]** until Phase-3 training completes; no synthetic-data number will be promoted into the descriptor.

3 Data Records

Records are deposited at [\[TODO: figshare/Zenodo DOI on deposit\]](#), with integrity verified by the per-file SHA-256 MANIFEST.sha256 at the archive root (an optional content-addressed mirror is noted in Section 6); their licensing follows the per-source basis set out in *Licensing and redistribution* below. The deposit is a single archive with the directory layout below; every leaf file is additionally listed with its SHA-256 in MANIFEST.sha256 at the archive root.

Licensing and redistribution. The corpus inherits the per-source redistribution basis of the WS-1 benchmark (to be confirmed, 2027a), which redistributes value-added derivatives only where the source licence permits and otherwise has reusers regenerate them locally. Accordingly: (i) all *numerical* records—credentials, `scan_meta.json`, `dataset_metadata.json`, and the integrity manifest—are released under CC BY 4.0, being aggregate statistics and metadata rather than image derivatives and carrying no protected health information; (ii) *pixel* records (reconstruction, uncertainty, error, and task-score maps) derived from LIDC-IDRI scans are released under CC BY 3.0, matching the licence under which WS-1 already redistributes its LIDC-derived simulated low-dose images; and (iii) pixel records derived from the AAPM 2016 Low-Dose CT Grand Challenge and Mayo LDCT-PD scans are *not* redistributed—those source data-use agreements restrict redistribution of pixel derivatives—and are instead regenerated locally by reusers from their own authorised copies of the source data using the deposited Apache-2.0 RunBundle (section 6), which reproduces them within floating-point tolerance. The released cross-vendor *credentials* and fidelity statistics (Table 6) are unaffected by (iii), as they are aggregate numbers rather than redistributed pixels. Generating code is Apache 2.0.

Record type	Format	Per-record content
Reference reconstruction	NIFTI (.nii.gz)	Ensemble-mean reconstruction $\hat{x}(y)$, HU, $512 \times 512 \times Z$
Uncertainty map	NIFTI	Per-pixel ensemble std. dev. $\sigma(y)$, same grid
Error map	NIFTI	Per-pixel $ \hat{x} - x^{\text{ref}} $ vs full dose, same grid
Task-score map	NIFTI	Frozen lung-nodule detector score map on $\hat{x}(y)$
Credential	JSON	5-tuple signal-equivalence credential (to be confirmed, 2027b) + verdict + CI + framework hash
Baseline reconstructions	NIFTI	Same fields, for 5 baseline methods (Sec. 3.1)
Per-scan metadata	JSON	Vendor, dose ratio r , anatomy, split, source-scan ID (links to WS-1)

Table 1: Overview of released record types. Counts and total size per type are in Table 4; the full directory manifest is below.

[TODO: FIGURE: Example released records for one test scan at $r = 0.25$, all on the identical voxel grid: (a) reference reconstruction $\hat{x}$ (HU); (b) per-pixel ensemble uncertainty σ (HU); (c) per-pixel error $e = |\hat{x} - x^{\text{ref}}|$ against the full-dose reference (HU); (d) frozen lung-nodule task-score map (unitless, $[0, 1]$). Illustrates the voxel-alignment guarantee and the paired (σ, e) calibration benchmark that is the corpus’s central reuse value. Populated at deposit (D9+270).]

Figure 1: Example released records for one test scan at $r = 0.25$, all on the identical voxel grid: (a) reference reconstruction $\hat{x}$ (HU); (b) per-pixel ensemble uncertainty σ (HU); (c) per-pixel error $e = |\hat{x} - x^{\text{ref}}|$ against the full-dose reference (HU); (d) frozen lung-nodule task-score map (unitless, $[0, 1]$). Illustrates the voxel-alignment guarantee and the paired (σ, e) calibration benchmark that is the corpus’s central reuse value. Populated at deposit (D9+270). (placeholder; populate when the corpus is deposited).

Directory layout.

```

pwm-ldct-recon-corpus/
  MANIFEST.sha256          # SHA-256 of every leaf file
  dataset_metadata.json    # ISA-style top-level metadata, DOI, licence
  reconstructions/
    <vendor>/<scan_id>/r<dd>/ # dd in {010,025,050,100}
      recon_mean.nii.gz      # reference reconstruction (this method)
      uncertainty_sigma.nii.gz # per-pixel ensemble std dev
      error_abs.nii.gz       # |recon - full_dose_ref|
      task_nodule_score.nii.gz # frozen-detector score map
      scan_meta.json         # vendor, r, anatomy, split, WS-1 source id
  baselines/
    <method>/<vendor>/<scan_id>/r<dd>/ # method in {fbp,redcnn,wgan_vgg,
      recon.nii.gz                  #               transformer,diffusion}
      error_abs.nii.gz
      task_nodule_score.nii.gz
  credentials/                  # cohort-level: one credential per
    <task>/r<dd>/<method>__<vendor>.json # (task x dose x method x vendor)
    all_credentials.jsonl       # flat queryable index of every credential
    checksums_and_versions.json # library version pins for regeneration

```

Naming and indexing. `scan_id` is the WS-1 source identifier, so any record back-links to its raw input. `r<dd>` encodes the dose ratio. The flat `all_credentials.jsonl` index lets a reuser query credentials (by vendor, dose, task, verdict) without walking the tree.

File-format conventions. All image records are single-file gzip-compressed NIfTI-1 (`.nii.gz`), `float32`, with an `sform` set to a scaled identity and voxel spacing carried in the header `pixdim`. The affine is an orientation-normalised scaled identity by design (patient-position and origin fields from the source DICOM are not propagated, as part of de-identification); records are therefore in a canonical voxel frame, not the original scanner frame, and should not be re-registered to source-patient coordinates. Reconstruction, reference, and error volumes are in Hounsfield units (HU); uncertainty maps are in HU (the ensemble standard deviation of the reconstruction); task-score maps are unitless detector scores in $[0, 1]$. All maps for one record share an identical voxel grid, so $\hat{x}$, σ , and e are voxel-aligned and can be stacked without resampling. JSON records are UTF-8; credentials validate against the machine-readable schema bundled with the deposit (`credential_schema.json`) and the WS-2 library ([to be confirmed, 2027b](#)).

Data dictionary. Each `scan_meta.json` carries the fields in [Table 2](#); each `credential.json` is the WS-2 5-tuple wire format (top-level `schema_version`, `framework_hash`, and a `credential` object holding `signal_ratio`, `task`, `epsilon`, `alpha`, `subpopulation`, `estimator`, `n_test`, `delta_mean`, `delta_ci_low`, `delta_ci_high`, `verdict`, and a `sample_size_check` block), specified field-by-field in the companion credential reading guide and the bundled JSON Schema.

Field	Type	Meaning
<code>scan_id</code>	string	WS-1 pseudonymous source identifier (back-link)
<code>vendor</code>	string	CT manufacturer (e.g. Siemens / GE / Canon)
<code>r</code>	float	signal ratio $\in (0, 1]$ (dose level)
<code>anatomy</code>	string	body region (<code>chest</code> in the v1 lung-nodule corpus)
<code>split</code>	string	always <code>test</code> for released records
<code>source_scan_id</code>	string	WS-1 series this record was reconstructed from
<code>low_dose_origin</code>	string	<code>real_paired</code> or <code>simulated</code> (see Methods)

Table 2: `scan_meta.json` fields. No field finer than anatomy/vendor is released (see Ethics and human-subjects provenance).

Vendor	Patients	Scans	Real-paired strata	Simulated strata
[TODO: vendor 1]	[TODO: N]	[TODO: N]	[TODO: N]	[TODO: N]
[TODO: vendor 2]	[TODO: N]	[TODO: N]	[TODO: N]	[TODO: N]
Total	[TODO: N]	[TODO: N]	[TODO: N]	[TODO: N]

Table 3: Cohort composition by vendor: distinct patients and scans, and how many (vendor, dose) strata are real-paired vs simulated low-dose. Populated at deposit (`package_corpus.py` derives the patient / scan counts from `scan_meta.json`).

Record type	Count	Format	Total size
Reference reconstructions	[TODO: N]	NIFTI	[TODO: GB]
Uncertainty maps	[TODO: N]	NIFTI	[TODO: GB]
Error maps	[TODO: N]	NIFTI	[TODO: GB]
Task-score maps	[TODO: N]	NIFTI	[TODO: GB]
Credentials	[TODO: N]	JSON	[TODO: MB]
Baseline reconstructions	[TODO: N]	NIFTI	[TODO: GB]
Total	[TODO: N]	—	[TODO: GB]

Table 4: Record counts and sizes. Populated when the corpus is generated and deposited (WS-3 Phase 3, D9+270).

3.1 Baseline reconstructions included for comparison

To make the corpus a self-contained comparison resource, the matched reconstructions of five baselines are included, each run through the identical emit path: FBP; RED-CNN (Chen et al., 2017); WGAN-VGG (Yang et al., 2018); a transformer reconstruction (Wang et al., 2023); and a diffusion / score-based reconstruction (Song et al., 2022). Each baseline is reproduced within ≤ 0.5 dB PSNR of its published numbers before its records are emitted; reproduction recipes accompany the deposit. The baselines are single-model methods and therefore carry no `uncertainty_sigma` record (ensemble disagreement is defined only for the deep-ensemble reference method); their record set is reconstruction, error map, task-score map, and credential.

4 Technical Validation

Validation establishes that the released records are fit for reuse. It is organised around the three properties a reuser depends on: that the reconstructions are faithful, that they are consistent across vendors, and that the released uncertainty maps actually track the released error maps.

Record integrity (construction check). As a reuse trust anchor we verify that every released `error_abs` map equals its defining quantity $|\hat{x}(y) - x^{\text{ref}}|$ by reconstruction rather than by assertion. The deposited tool `package_corpus.py verify-error-maps` draws a random sample of records, reloads the reconstruction together with the scan’s full-dose reference (the released $r = 1.00$ `recon_mean`), recomputes $|\hat{x} - x^{\text{ref}}|$, and confirms it matches the released `error_abs` voxelwise within float32 tolerance ($\leq 10^{-3}$ HU). The synthetic end-to-end fixture runs this check in continuous integration on every commit, and a reuser can re-run it on the deposit with a single command, so the error layer is independently checkable, not merely trusted.

Reconstruction fidelity. Table 5 reports PSNR, SSIM, and LPIPS (Zhang et al., 2018) of every released reconstruction against its full-dose reference, per dose level, for the reference method and the five baselines. These quantify how close each released volume is to ground truth; they are descriptive data-quality evidence, not a competition.

Method	$r = 0.10$		$r = 0.25$		$r = 0.50$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
FBP	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RED-CNN	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
WGAN-VGG	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Transformer	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Diffusion	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Reference	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 5: Reconstruction fidelity of the released records against the full-dose reference, with 95% bootstrap CIs ($B=1000$) over test patients. Populated at deposit (D9+270).

Cross-vendor consistency: interpolation vs. extrapolation. Because every record is vendor-labelled, reusers can study cross-vendor generalisation directly — but only if the corpus states which vendors are *interpolative* (comfortably inside the multi-vendor training distribution) and which are *extrapolative* (at its edge). We make this operational with leave-one-vendor-out (LOVO) generation. The released reference reconstructions are produced by the deployed ensemble trained on *all* vendors, so for the deposited corpus every vendor is in-distribution; Table 6 additionally reconstructs each vendor’s scans with an ensemble trained *without* that vendor and reports the fidelity drop $\Delta = \text{PSNR}_{\text{in-dist}} - \text{PSNR}_{\text{held-out}}$. Δ is the empirical distance of a vendor from the rest: a small Δ marks an **interpolation** vendor whose reconstructions a cross-vendor model is expected to reproduce, while a large Δ marks an **extrapolation** vendor whose records should anchor out-of-distribution claims and be scoped cautiously. Each vendor’s regime is labelled in the table from its Δ against a deposit-fixed threshold, so a reuser can pick interpolative vendors for in-distribution benchmarks and extrapolative vendors for robustness stress-tests without re-deriving anything. The Δ values themselves are populated from the Phase-3 runs (Table 5’s generation), so this is the framing under which those numbers are read, not a separate experiment.

Held-out vendor	In-dist. PSNR	Held-out PSNR	Δ (dB)	Regime
[TODO: vendor 1]	[TODO:]	[TODO:]	[TODO:]	[TODO: interp/extrap]
[TODO: vendor 2]	[TODO:]	[TODO:]	[TODO:]	[TODO: interp/extrap]
[TODO: vendor 3]	[TODO:]	[TODO:]	[TODO:]	[TODO: interp/extrap]

Table 6: Cross-vendor fidelity under leave-one-vendor-out generation: PSNR of each vendor’s scans reconstructed by the all-vendor ensemble (in-distribution) versus an ensemble trained without that vendor (held-out), and the drop Δ . The per-vendor **Regime** label (*interpolation* if Δ is below the deposit-fixed threshold, else *extrapolation*) tells reusers which released vendor records are safe for in-distribution reuse versus out-of-distribution stress-tests. Populated at deposit (WS-3 Phase 3, D9+270).

Uncertainty–error agreement (central reuse validation). The corpus’s distinctive reuse value is the paired (uncertainty, error) records, so we validate it along three complementary axes rather than one. **(i) Rank tracking:** the Spearman correlation between $\sigma(y)$ and $e(y)$, per pixel and aggregated per case (Table 7) — does larger predicted uncertainty co-occur with larger error? **(ii) Calibration magnitude:** a reliability diagram of predicted σ against observed RMS error per σ -bin (Figure 2), summarised by an expected-calibration-error (ECE), the mean absolute gap between the two — rank correlation can be high while the variance is systematically mis-scaled, which ECE catches. **(iii) Interval coverage:** the empirical fraction of voxels whose true error falls within $\pm k \sigma$ for $k \in \{1, 2, 3\}$ (Table 8); a well-calibrated Gaussian-like variance yields $\approx 68/95/99.7\%$. Reporting all three lets a reuser distinguish a uncertainty map that merely *looks* error from one that is *quantitatively* calibrated for interval-based triage; where they disagree, the corpus documents which property holds so the variance is not over-trusted. This is the single most important validation for the corpus’s stated calibration-research reuse.

Dose	Spearman ρ (per-pixel)	Spearman ρ (per-case mean)	95% CI
$r = 0.10$	[TODO:]	[TODO:]	[TODO:]
$r = 0.25$	[TODO:]	[TODO:]	[TODO:]
$r = 0.50$	[TODO:]	[TODO:]	[TODO:]

Table 7: Agreement between released uncertainty maps and released error maps. Populated at deposit (D9+270).

[TODO: FIGURE: Reliability diagram: released per-pixel ensemble standard deviation vs observed per-pixel reconstruction error (RMS per σ -bin) on the held-out test split, annotated with the expected-calibration-error (ECE).]

Figure 2: Reliability diagram: released per-pixel ensemble standard deviation vs observed per-pixel reconstruction error (RMS per σ -bin) on the held-out test split, annotated with the expected-calibration-error (ECE). (placeholder; populate when the corpus is deposited).

Downstream-task validity. Table 9 reports lung-nodule detection AUC (fixed FPR = 0.1) computed from the released task-score maps with a frozen detector held constant across all methods. This validates that the released reconstructions preserve diagnostically relevant structure, and documents the known divergence between pixel fidelity and task performance (Kim et al., 2019) so reusers interpret both layers correctly.

Credential validation. Each released credential is internally audited with the `pwm-audit` tool (to be confirmed, 2027b) (schema validity, verdict-vs-CI self-consistency, framework-hash recognition); Table 10 summarises the verdicts. The audit is bundled in the deposit so a reuser can re-verify every credential without re-running the bootstrap.

Reproducibility of generation. The full corpus regenerates from the deposited Apache-2.0 RunBundle: `docker run` on a fresh machine reproduces every record bit-identically within IEEE-754 tolerance (seeds, library pins, and CuDNN-deterministic settings in Supplementary Table S2). The RunBundle is independently re-executed by the PWM challenge scoring service before posting, providing third-party regeneration evidence.

5 Usage Notes

Loading. All image records are NIfTI and load with any standard reader (`nibabel`, ITK); credentials are JSON validated against the published schema (to be confirmed, 2027b). The `all_credentials.jsonl` index supports tabular queries by vendor, dose, task, and verdict. Each record’s `scan_meta.json` links back to its WS-1 raw source (to be confirmed, 2027a). A minimal read of one record’s voxel-aligned reconstruction, uncertainty, and error maps:

```
import nibabel as nib, json
base = "reconstructions/Siemens/<scan_id>/r025/"
recon = nib.load(base + "recon_mean.nii.gz").get_fdata()      # HU
sigma = nib.load(base + "uncertainty_sigma.nii.gz").get_fdata() # HU, same grid
error = nib.load(base + "error_abs.nii.gz").get_fdata()       # |recon - full_dose|
meta = json.load(open(base + "scan_meta.json"))               # vendor, r, anatomy, ...
# the (sigma, error) pair is the ready-made calibration benchmark
```

Intended reuses. (1) Uncertainty-calibration research: treat the paired (`uncertainty_sigma`, `error_abs`) maps as a ready calibration benchmark. (2) Cross-vendor robustness: use vendor labels and the leave-one-vendor-out validation to scope claims. (3) Dose-dependent downstream-task studies: use the released task-score maps without re-deriving reconstructions. (4) Method comparison: benchmark a new reconstruction against the matched baseline records on identical inputs and report a credential against the same framework hash.

Dose	ECE	$\pm 1\sigma$ cov.	$\pm 2\sigma$ cov.	$\pm 3\sigma$ cov.
$r = 0.10$	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$r = 0.25$	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$r = 0.50$	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 8: Calibration magnitude (expected-calibration-error) and empirical $\pm k\sigma$ interval coverage of the released uncertainty maps; well-calibrated variance yields coverage $\approx 68/95/99.7\%$. Populated at deposit (D9+270).

[TODO: TABLE: Lung-nodule detection AUC (frozen detector, fixed FPR = 0.1) computed from the released task-score maps, per method, per dose level]

Table 9: Lung-nodule detection AUC (frozen detector, fixed FPR = 0.1) computed from the released task-score maps, per method, per dose level (placeholder; populate when the corpus is deposited).

Cautions for reuse. The released uncertainty is ensemble disagreement; it can be non-informative where all members fail in correlated ways (out-of-distribution anatomy)—inspect the reliability diagram before triaging on σ . Cross-vendor records for vendors not in training are extrapolation, not measurement. Dose ratios below 0.10 are not characterised. Reconstructions are single-slice 2D. Credential **INDETERMINATE** verdicts on this cohort reflect sample size (to be confirmed, 2027b), not evidence of non-equivalence, and should not be read as a negative finding about a method.

Licensing. Numerical records (credentials, metadata, manifest): CC BY 4.0. LIDC-derived pixel records: CC BY 3.0. Pixel records derived from the DUA-restricted AAPM 2016 and Mayo LDCT-PD scans are not redistributed and are regenerated locally via the RunBundle (see *Licensing and redistribution* under Data Records, section 3). Generating code and RunBundle: Apache 2.0. Reuse requires citing this Data Descriptor and the WS-1 source dataset (to be confirmed, 2027a).

6 Code Availability

The complete generating pipeline—training, inference, ensemble aggregation, error/task/credential emission, and the evaluation harness that produced every Technical Validation number—is released under Apache 2.0 at https://github.com/integritynoble/low_dose_CT/tree/main/WS-3_reference_method, and packaged as a pinned Docker RunBundle; integrity is provided by the deposited SHA-256 MANIFEST.sha256, with an optional content-addressed mirror (IPFS CID [TODO: at deposit]) as a supplementary check. Pre-trained ensemble weights, the pinned `Dockerfile`, evaluation scripts, random seeds, and the training-compute disclosure (Supplementary Table S3) permit bit-identical regeneration of the corpus within floating-point tolerance on equivalent hardware. The forward model ships in the pipeline as a self-contained parallel-beam Radon implementation, interface-compatible with (but not dependent on) `pwm_core.contrib.modalities.ct_radon` (PWM Protocol, 2026); the credentialing library is `pwm_dose_equivalence` (to be confirmed, 2027b). The credential-emission and corpus-packaging steps (record layout, `pwm-audited` credential issuance, machine-readable metadata, and the SHA-256 integrity manifest) are unit-tested, and a runnable synthetic fixture reproduces the full record-generation $\rightarrow$ credential $\rightarrow$ metadata $\rightarrow$ manifest pipeline end-to-end on miniature data, so a reviewer can exercise the corpus structure and audit chain in one command without the full dataset.

Data Citations

The deposited corpus: [TODO: figshare/Zenodo DOI on deposit] (CC BY 4.0).

Source benchmark dataset (WS-1): to be confirmed (2027a).

Signal-equivalence framework and credential schema: to be confirmed (2027b).

r	Task	ε	α	Subpopulation	Verdict	CI
0.25	Lung nodule AUC	0.05	0.05	adult-chest test split	[TODO:]	[TODO:]
0.10	Lung nodule AUC	0.05	0.05	adult-chest test split	[TODO:]	[TODO:]

Table 10: Signal-equivalence credentials (to be confirmed, 2027b) for the reference reconstruction against full-dose, audited with `pwm-audit`. Verdicts: PASS / FAIL / INDETERMINATE. Note: the WS-1 test cohort sits in the INDETERMINATE-dominated regime for the AUC default; absence of PASS reflects cohort size, not non-equivalence (see to be confirmed, 2027b).

Acknowledgments

[TODO: Acknowledge compute providers and the source-data custodians (Mayo Clinic / AAPM Low-Dose CT Grand Challenge for controlled-access scans; TCIA/LIDC-IDRI). Fill at submission.]

Author Contributions

[TODO: State per the CRediT taxonomy. Template: “A.A. and B.B. designed the reconstruction pipeline and generated the corpus; A.A. performed the technical validation; C.C. computed and audited the signal-equivalence credentials; A.A. drafted the manuscript; all authors reviewed and approved the final manuscript.” Every listed author must have a contribution.]

Competing Interests

[TODO: Declare all financial *and* non-financial competing interests (*Scientific Data* requires both). In particular, disclose the PWM Protocol Foundation affiliation. Template if none material: “The authors declare no competing interests.”]

Funding

[TODO: List funders and grant numbers (e.g. NIH R21, if awarded — see WS-5), or state “This work received no specific grant from any funding agency.” May be merged into Acknowledgments at submission.]

References

- AAPM. 2016 NIH-AAPM-Mayo clinic low dose CT grand challenge. <https://www.aapm.org/grandchallenge/lowdosect/>, 2016.
- Jonas Adler and Ozan Öktem. Learned primal-dual reconstruction. *IEEE Transactions on Medical Imaging*, 37(6):1322–1332, 2018.
- S. G. Armato III, G. McLennan, L. Bidaut, et al. The LIDC/IDRI database. *Medical Physics* 38(2):915-931, 2011.
- Michael Baumgartner, Paul F. Jäger, Fabian Isensee, and Klaus H. Maier-Hein. nnDetection: A self-configuring method for medical object detection. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 530–539, 2021.
- Hu Chen, Yi Zhang, Mannudeep K. Kalra, et al. Low-dose CT with a residual encoder-decoder convolutional neural network. *IEEE Transactions on Medical Imaging*, 36(12):2524–2535, 2017.

- Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, 2016.
- H. Kim et al. Diagnostic accuracy of low-dose CT with iterative reconstruction. *European Radiology*, 29: 3146–3155, 2019.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- Morteza Mardani, Qingyun Sun, David Donoho, Vardan Papyan, Hatef Monajemi, Shreyas Vasanawala, and John Pauly. Neural proximal gradient descent for compressive imaging. *Advances in Neural Information Processing Systems*, 2018.
- Cynthia H. McCollough, Adam C. Bartley, Rickey E. Carter, et al. Low-dose CT for the detection and classification of metastatic liver lesions: Results of the 2016 low dose CT grand challenge. *Medical Physics*, 44(10):e339–e352, 2017.
- Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, 2019.
- PWM Protocol. PWM Core: A 172-modality engine for verified scientific computing. <https://github.com/integritynoble/pwm>, 2026.
- Arnaud A. A. Setio, Alberto Traverso, Thomas de Bel, et al. Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge. *Medical Image Analysis*, 42:1–13, 2017.
- Yang Song, Liye Shen, Lei Xing, and Stefano Ermon. Solving inverse problems in medical imaging with score-based generative models. In *International Conference on Learning Representations (ICLR)*, 2022.
- Authors to be confirmed. PWM-LDCT 1.0: A content-addressed, multi-task harmonization of three public low-dose CT datasets. Manuscript in preparation, target *Nature Scientific Data*, 2027a.
- Authors to be confirmed. Signal-equivalence: A probabilistic framework for comparing reconstruction methods across reduced-signal imaging modalities. Manuscript in preparation, target *Nature Methods*, 2027b.
- Dayang Wang, Fenglei Fan, Zhan Wu, Rui Liu, Fei Wang, and Hengyong Yu. CTformer: Convolution-free token2token dilated vision transformer for low-dose CT denoising. *Physics in Medicine and Biology*, 68(6): 065012, 2023.
- Ge Wang, Jong Chul Ye, and Bruno De Man. Deep learning for tomographic image reconstruction. *Nature Machine Intelligence*, 2(12):737–748, 2020.
- Qingsong Yang, Pingkun Yan, Yanbo Zhang, et al. Low-dose CT image denoising using a generative adversarial network with Wasserstein distance and perceptual loss. *IEEE Transactions on Medical Imaging*, 37(6): 1348–1357, 2018.
- Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.